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JEE Main Physics Practice Sheet — DPP

Topic: Centre of Mass of a Two-Particle System

Time Allowed: 60 Minutes — **Max Marks:** 80 — **Marking Scheme:** +4, -1

Q1. Two particles of masses $m_1 = 2$ kg and $m_2 = 3$ kg are located at positions $\vec{r}_1 = (2\hat{i} + 4\hat{j})$ m and $\vec{r}_2 = (7\hat{i} - \hat{j})$ m respectively. The position vector of their centre of mass is:

(A) $(5\hat{i} + \hat{j})$ m

(B) $(4\hat{i} + 2\hat{j})$ m

(C) $(9\hat{i} + 3\hat{j})$ m

(D) $(5\hat{i} + 2\hat{j})$ m

Q2. Two particles of masses m_1 and m_2 are separated by a fixed distance d. The distance of the centre of mass from the particle of mass m_1 is:

(A) $\dfrac{m_1 d}{m_1 + m_2}$

(B) $\dfrac{(m_1 + m_2)d}{m_1}$

(C) $\dfrac{m_2 d}{m_1 + m_2}$

(D) $\dfrac{m_2 d}{m_1}$

Q3. Two bodies of masses 2 kg and 4 kg are moving along the x-axis with velocities 6 m/s and -3 m/s respectively. The velocity of their centre of mass is:

(A) 2 m/s

(B) -1 m/s

(C) 1 m/s

(D) 0 m/s

Q4. Two particles of masses $m_1 = 3$ kg and $m_2 = 1$ kg are initially at rest at a separation of 12 m. They start moving towards each other under mutual gravitational attraction. The displacement of mass m_1 when the two particles collide is:

(A) 3 m

(B) 4 m

(C) 9 m

(D) 6 m

Q5. In a two-particle system with masses m_1 and m_2, the sum of moments of the masses about their centre of mass $(\sum m_i \vec{r}_i')$ is always:

(A) Dependent on the external forces

(B) Non-zero and positive

(C) Dependent on the choice of origin

(D) Strictly equal to zero

Q6. Two masses $m_1 = 4$ kg and $m_2 = 6$ kg are separated by a distance L. If the mass m_1 is displaced towards the right by a distance of 5 cm, by what distance and in which direction must mass m_2 be shifted so that the position of the centre of mass remains unchanged?

(A) $\dfrac{10}{3}$ cm towards left

(B) 5 cm towards left

(C) $\dfrac{10}{3}$ cm towards right

(D) $\dfrac{15}{2}$ cm towards right

Q7. Two masses m_1 and m_2 are connected by a light spring of spring constant k and placed on a smooth horizontal surface. Initially at rest, they are pulled apart and released. If at an instant the acceleration of m_1 is a_1 towards right, the acceleration of the centre of mass of the two-particle system is:

(A) $\dfrac{m_1 a_1}{m_1 + m_2}$

(B) 0

(C) $\dfrac{m_2 a_1}{m_1}$

(D) a_1

Q8. A system consists of two particles of masses 1 kg and 2 kg. The first particle has an acceleration $\vec{a}_1 = (2\hat{i} + 3\hat{j})$ m/s^2 and the second particle has an acceleration $\vec{a}_2 = (-\hat{i} + 6\hat{j})$ m/s^2. The magnitude of the acceleration of the centre of mass is:

(A) 5 m/s^2

(B) $\sqrt{26}$ m/s^2

(C) 3 m/s^2

(D) 4 m/s^2

Q9. If the internal forces acting between the two particles of an isolated system are non-zero, the motion of their centre of mass:

(A) Must be accelerated

(B) Follows a curved trajectory

(C) Is completely unaffected by internal forces

(D) Depends on the magnitude of internal forces

Q10. Two particles of masses $m_1 = 1$ kg and $m_2 = 3$ kg have position vectors $\vec{r}_1(t) = (2t^2\hat{i} + 3t\hat{j})$ m and $\vec{r}_2(t) = (t\hat{i} + 2t^2\hat{j})$ m respectively. The total external force acting on the two-particle system at any time t is:

(A) $(2\hat{i} + 6\hat{j})$ N

(B) $(4\hat{i} + 12\hat{j})$ N

(C) $(4\hat{i} + 6\hat{j})$ N

(D) $(8\hat{i} + 12\hat{j})$ N

Q11. In a diatomic molecule of carbon monoxide (CO), the distance between the carbon atom ($m_C = 12$ u) and the oxygen atom ($m_O = 16$ u) is 1.12 Å. The distance of the centre of mass from the carbon atom is:

(A) 0.48 Å
(B) 0.56 Å
(C) 0.64 Å
(D) 0.72 Å

Q12. Two particles of masses $m_1 = 2$ kg and $m_2 = 3$ kg are projected vertically upward with speeds 30 m/s and 20 m/s respectively from the ground at $t = 0$. Neglecting air resistance, the maximum vertical height reached by the centre of mass of the two particles is ($g = 10$ m/s^2):

(A) 36.0 m
(B) 28.8 m
(C) 32.4 m
(D) 25.0 m

Q13. A man of mass 50 kg stands at one end of a stationary wooden plank of mass 150 kg and length 4 m floating on calm water. If the man walks to the other end of the plank, the displacement of the plank relative to the water is:

(A) 1.0 m
(B) 1.33 m
(C) 0.75 m
(D) 2.0 m

Q14. The kinetic energy of a two-particle system of masses m_1 and m_2 with velocities $\vec{v}_1$ and $\vec{v}_2$ can be expressed as $K = \frac{1}{2}(m_1 + m_2)v_{\text{cm}}^2 + K_{\text{rel}}$. The internal kinetic energy K_{rel} relative to the centre of mass is:

(A) $\frac{1}{2}\left(\frac{m_1 m_2}{m_1 + m_2}\right)|\vec{v}_1 + \vec{v}_2|^2$
(B) $\frac{1}{2}(m_1 + m_2)|\vec{v}_1 - \vec{v}_2|^2$
(C) $\frac{1}{2}\left(\frac{m_1 m_2}{m_1 + m_2}\right)|\vec{v}_1 - \vec{v}_2|^2$
(D) $\frac{1}{2}\sqrt{m_1 m_2}|\vec{v}_1 - \vec{v}_2|^2$

Q15. Two particles of masses $m_1 = 2$ kg and $m_2 = 4$ kg are moving towards each other along the x-axis. In the centre of mass (C-frame) reference frame, the ratio of the magnitude of linear momentum of m_1 to that of m_2 ($p_{1,\text{cm}} : p_{2,\text{cm}}$) is:

(A) $1:2$
(B) $2:1$
(C) $1:1$
(D) $1:4$

Q16. Two particles of masses 2 kg and 3 kg are located at coordinates $(1, 2)$ and $(6, 7)$ respectively. If the origin of the coordinate system is shifted to $(2, 3)$, the coordinates of their centre of mass in the new reference frame are:

(A) $(2, 2)$
(B) $(4, 5)$
(C) $(3, 3)$
(D) $(1, 2)$

Q17. Two particles of equal mass m are projected simultaneously from the ground with the same initial speed u at angles 30° and 60° with the horizontal in the same vertical plane. The trajectory traced by their centre of mass before either particle lands is:

(A) A vertical straight line
(B) A parabola
(C) A circular arc
(D) An inclined straight line

Q18. Two particles of masses m_1 and m_2 have initial velocities $\vec{u}_1$ and $\vec{u}_2$. A constant external force $\vec{F}$ acts on m_1 while no external force acts on m_2. The velocity of the centre of mass after time t is:

(A) $\frac{m_1\vec{u}_1 + m_2\vec{u}_2}{m_1 + m_2} + \frac{\vec{F}t}{m_1}$
(B) $\frac{m_1\vec{u}_1 + m_2\vec{u}_2 + \vec{F}t}{m_1 + m_2}$
(C) $\frac{\vec{u}_1 + \vec{u}_2}{2} + \frac{\vec{F}t}{m_1 + m_2}$
(D) $\frac{m_1\vec{u}_1 + m_2\vec{u}_2}{m_1 + m_2} + \frac{\vec{F}t}{m_2}$

Q19. A mass $m_1 = 5$ kg is located at $x = 0$ and another mass $m_2 = 15$ kg is located at $x = 20$ m. If mass m_1 is moved by 4 m along $+x$, the distance by which the centre of mass shifts is:

(A) 1.0 m
(B) 2.0 m
(C) 3.0 m
(D) 0.5 m

Q20. Two particles of masses m and $2m$ are connected by a massless rigid rod of length L. The moment of inertia of this two-particle system about an axis passing through its centre of mass and perpendicular to the rod is:

(A) $\frac{1}{3}mL^2$
(B) $\frac{3}{2}mL^2$
(C) $\frac{2}{3}mL^2$
(D) $\frac{2}{9}mL^2$

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Centre of Mass of a Two-Particle System

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	A	6	A	11	C	16	A
2	C	7	B	12	B	17	B
3	D	8	A	13	A	18	B
4	A	9	C	14	C	19	A
5	D	10	B	15	C	20	C

Detailed Step-by-Step Solutions

Sol 1. Option (A)
Position vector of the centre of mass:

$$\vec{r}_{\text{cm}} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1+m_2}$$

Substituting values:

$$\vec{r}_{\text{cm}} = \frac{2(2\hat{\imath}+4\hat{\jmath})+3(7\hat{\imath}-\hat{\jmath})}{2+3} = \frac{(4\hat{\imath}+8\hat{\jmath})+(21\hat{\imath}-3\hat{\jmath})}{5}$$

$$\vec{r}_{\text{cm}} = \frac{25\hat{\imath}+5\hat{\jmath}}{5} = (5\hat{\imath}+\hat{\jmath})\ \text{m}$$

Sol 2. Option (C)
Taking m_1 at the origin ($x_1 = 0$) and m_2 at $x_2 = d$:

$$x_{\text{cm}} = \frac{m_1(0)+m_2(d)}{m_1+m_2} = \frac{m_2 d}{m_1+m_2}$$

Sol 3. Option (D)
Velocity of the centre of mass:

$$v_{\text{cm}} = \frac{m_1v_1+m_2v_2}{m_1+m_2} = \frac{2(6)+4(-3)}{2+4} = \frac{12-12}{6} = 0\ \text{m/s}$$

Sol 4. Option (A)
Since no external horizontal force acts on the system, the centre of mass remains stationary ($\Delta x_{\text{cm}} = 0$).
Taking m_1 at $x = 0$ and m_2 at $x = 12$ m:

$$x_{\text{cm}} = \frac{3(0)+1(12)}{3+1} = \frac{12}{4} = 3\ \text{m}$$

Because both particles collide at the centre of mass position, the displacement of m_1 is:

$$\Delta x_1 = 3\ \text{m} - 0\ \text{m} = 3\ \text{m}$$

Sol 5. Option (D)
By definition of the centre of mass, the position vector of particle i relative to the centre of mass is $\vec{r}\,'_i = \vec{r}_i - \vec{r}_{\text{cm}}$.
Sum of mass moments:

$$\sum m_i\vec{r}\,'_i = \sum m_i(\vec{r}_i-\vec{r}_{\text{cm}}) = \sum m_i\vec{r}_i - \vec{r}_{\text{cm}}\sum m_i = M\vec{r}_{\text{cm}} - M\vec{r}_{\text{cm}} = \vec{0}$$

Hence, it is identically zero for all systems.

Sol 6. Option (A)
For the position of centre of mass to remain unchanged:

$$\Delta x_{\text{cm}} = \frac{m_1\Delta x_1+m_2\Delta x_2}{m_1+m_2} = 0 \implies m_1\Delta x_1 + m_2\Delta x_2 = 0$$

Given $m_1 = 4$ kg, $\Delta x_1 = +5$ cm, $m_2 = 6$ kg:

$$4(+5)+6(\Delta x_2) = 0 \implies \Delta x_2 = -\frac{20}{6} = -\frac{10}{3}\ \text{cm}$$

The negative sign indicates mass m_2 must be shifted towards the left by $\frac{10}{3}$ cm.

Sol 7. Option (B)
The spring force between m_1 and m_2 is an internal action-reaction force. Since the horizontal surface is smooth, the net external force on the system is zero ($\sum \vec{F}_{\text{ext}} = \vec{0}$). Therefore:

$$\vec{a}_{\text{cm}} = \frac{\sum \vec{F}_{\text{ext}}}{m_1+m_2} = 0$$

Sol 8. Option (A)
Acceleration of the centre of mass:

$$\vec{a}_{\text{cm}} = \frac{m_1\vec{a}_1+m_2\vec{a}_2}{m_1+m_2} = \frac{1(2\hat{\imath}+3\hat{\jmath})+2(-\hat{\imath}+6\hat{\jmath})}{1+2}$$

$$\vec{a}_{\text{cm}} = \frac{(2\hat{\imath}+3\hat{\jmath})+(-2\hat{\imath}+12\hat{\jmath})}{3} = \frac{0\hat{\imath}+15\hat{\jmath}}{3} = 5\hat{\jmath}\ \text{m/s}^2$$

Magnitude $|\vec{a}_{\text{cm}}| = 5\ \text{m/s}^2$.

Sol 9. Option (C)
According to Newton's third law, internal forces between particles in a system occur in equal and opposite pairs ($\vec{F}_{12} = -\vec{F}_{21}$). Their vector sum is zero, so internal forces have zero net contribution to $\vec{a}_{\text{cm}}$. Thus, the motion of the centre of mass is completely independent of internal forces.

Sol 10. Option (B)
Accelerations of the individual particles are:

$$\vec{a}_1 = \frac{d^2\vec{r}_1}{dt^2} = \frac{d^2}{dt^2}(2t^2\hat{\imath}+3t\hat{\jmath}) = 4\hat{\imath}\ \text{m/s}^2$$

$$\vec{a}_2 = \frac{d^2\vec{r}_2}{dt^2} = \frac{d^2}{dt^2}(t\hat{\imath}+2t^2\hat{\jmath}) = 4\hat{\jmath}\ \text{m/s}^2$$

The net external force on the system is:

$$\vec{F}_{\text{ext}} = m_1\vec{a}_1 + m_2\vec{a}_2 = 1(4\hat{\imath})+3(4\hat{\jmath}) = (4\hat{\imath}+12\hat{\jmath})\ \text{N}$$

Sol 11. Option (C)
Taking the carbon atom at the origin ($x_{\text{C}} = 0$) and oxygen atom at $x_{\text{O}} = 1.12$ Å:

$$x_{\text{cm}} = \frac{m_{\text{C}}(0)+m_{\text{O}}(1.12)}{m_{\text{C}}+m_{\text{O}}} = \frac{16\times1.12}{12+16} = \frac{16\times1.12}{28} = \frac{4\times1.12}{7} =$$

Sol 12. Option (B)
Initial vertical velocity of the centre of mass:

$$u_{\text{cm}} = \frac{m_1u_1 + m_2u_2}{m_1 + m_2} = \frac{2(30) + 3(20)}{2+3} = \frac{60+60}{5} = \frac{120}{5} = 24 \text{ m/s}$$

Since both particles experience uniform downward acceleration due to gravity $g = 10$ m/s^2, the centre of mass has constant downward acceleration $a_{\text{cm}} = -g$.
Maximum height attained by the centre of mass:

$$H_{\max} = \frac{u_{\text{cm}}^2}{2g} = \frac{24^2}{2\times 10} = \frac{576}{20} = 28.8 \text{ m}$$

Sol 13. Option (A)
Let the displacement of the plank relative to water be x in the opposite direction. Since no external horizontal force acts:

$$m_{\text{man}}(L-x) - m_{\text{plank}}(x) = 0 \implies x = \frac{m_{\text{man}}L}{m_{\text{man}} + m_{\text{plank}}}$$

Substituting given values:

$$x = \frac{50\times 4}{50+150} = \frac{200}{200} = 1.0 \text{ m}$$

Sol 14. Option (C)
By König's theorem for a two-body system, total kinetic energy is decomposed into the kinetic energy of the centre of mass and relative motion:

$$K = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}\mu v_{\text{rel}}^2$$

where $\mu = \frac{m_1m_2}{m_1+m_2}$ is the reduced mass and $\vec{v}_{\text{rel}} = \vec{v}_1 - \vec{v}_2$.
Thus, $K_{\text{rel}} = \frac{1}{2}\left(\frac{m_1m_2}{m_1+m_2}\right)|\vec{v}_1 - \vec{v}_2|^2$.

Sol 15. Option (C)
In the centre of mass reference frame, total momentum is identically zero by definition:

$$\vec{P}_{\text{cm}} = \vec{p}_{1,\text{cm}} + \vec{p}_{2,\text{cm}} = \vec{0} \implies \vec{p}_{1,\text{cm}} = -\vec{p}_{2,\text{cm}}$$

Therefore, the magnitudes are always strictly equal:

$$|\vec{p}_{1,\text{cm}}| = |\vec{p}_{2,\text{cm}}| \implies \text{Ratio } = 1:1$$

Sol 16. Option (A)
In the original frame:

$$x_{\text{cm}} = \frac{2(1)+3(6)}{2+3} = \frac{20}{5} = 4$$

$$y_{\text{cm}} = \frac{2(2)+3(7)}{2+3} = \frac{25}{5} = 5$$

When the origin is shifted to $(x_0, y_0) = (2, 3)$:

$$x'_{\text{cm}} = x_{\text{cm}} - x_0 = 4 - 2 = 2$$

$$y'_{\text{cm}} = y_{\text{cm}} - y_0 = 5 - 3 = 2$$

So the coordinates in the new frame are $(2, 2)$.

Sol 17. Option (B)
Both particles move under gravity with identical downward acceleration $\vec{g} = -g\hat{j}$.
Thus, the acceleration of the centre of mass is $\vec{a}_{\text{cm}} = \vec{g}$.
The initial velocity of COM is:

$$\vec{u}_{\text{cm}} = \frac{u(\cos 30^\circ\hat{i} + \sin 30^\circ\hat{j}) + u(\cos 60^\circ\hat{i} + \sin 60^\circ\hat{j})}{2}$$

Since $\vec{u}_{\text{cm}}$ has non-zero horizontal and vertical components while experiencing constant vertical downward acceleration $\vec{g}$, its trajectory is a parabola.

Sol 18. Option (B)
Initial velocity of COM: $\vec{v}_{\text{cm}}(0) = \frac{m_1\vec{u}_1 + m_2\vec{u}_2}{m_1+m_2}$.
Acceleration of COM: $\vec{a}_{\text{cm}} = \frac{\sum \vec{F}_{\text{ext}}}{m_1+m_2} = \frac{\vec{F}}{m_1+m_2}$.
Using $\vec{v}_{\text{cm}}(t) = \vec{v}_{\text{cm}}(0) + \vec{a}_{\text{cm}}t$:

$$\vec{v}_{\text{cm}}(t) = \frac{m_1\vec{u}_1 + m_2\vec{u}_2 + \vec{F}t}{m_1+m_2}$$

Sol 19. Option (A)
Shift in the position of the centre of mass:

$$\Delta x_{\text{cm}} = \frac{m_1\Delta x_1 + m_2\Delta x_2}{m_1+m_2} = \frac{5(4)+15(0)}{5+15} = \frac{20}{20} = 1.0 \text{ m}$$

Sol 20. Option (C)
Distance of m from COM: $r_1 = \frac{2m}{m+2m}L = \frac{2}{3}L$.
Distance of $2m$ from COM: $r_2 = \frac{m}{m+2m}L = \frac{1}{3}L$.
Moment of inertia about the COM axis:

$$I = mr_1^2 + (2m)r_2^2 = m\left(\frac{2}{3}L\right)^2 + 2m\left(\frac{1}{3}L\right)^2 = \frac{4}{9}mL^2 + \frac{2}{9}mL^2 = \frac{6}{9}$$